ENVS-193DS Final

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Link to repo

```r
# read in packages
library(tidyverse)
library(here)
library(DHARMa)
library(ggeffects)
library(rstatix)
library(janitor)

# reading in data and creating new object
nest <- read_csv(
  here("data", "nest_data_final.csv"))

# creating new data frame with personal data
personal_data <- read_csv(here("data", "personal-datasheet.csv")) |>
  # standardizing column names
  clean_names() |>
  # dropping rows with na observations
  drop_na() |>
  # changing the stop column to display TRUE/FALSE instead of 1/0
  mutate(stop = as.logical(stop),
         week = as.factor(week))
```

Problem 1. Research writing

Problem 2. Data Analysis

a.

```r
# creating clean data frame
nests_clean <- nest |>
  # selecting desired columns
  select(ant_species, height_cm, case_control) |>
  # filtering for desired ant species
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
  # creating new column for full scientific name
  mutate(sci_name = case_when(
      ant_species == "AB" ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR"  ~ "Crematogaster nigriceps"),
```

```r
      # reordering factors for ant species
    ant_species = factor(ant_species,
      levels = c("RRB", "BBR", "AB"))
    )

# displaying structure of data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "RRB","BBR","AB": 1 1 1 1 3 1 2 1 3 2 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ sci_name    : chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster m
```

```r
# slicing sample
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   ant_species height_cm case_control sci_name
   <fct>           <dbl>        <dbl> <chr>
 1 BBR              201             1 Crematogaster nigriceps
 2 RRB              192             0 Crematogaster mimosae
 3 RRB              154.            0 Crematogaster mimosae
 4 RRB              406             0 Crematogaster mimosae
 5 RRB              157             0 Crematogaster mimosae
 6 RRB              144             0 Crematogaster mimosae
 7 BBR              132.            0 Crematogaster nigriceps
 8 RRB              134             0 Crematogaster mimosae
 9 RRB              236             0 Crematogaster mimosae
10 RRB              144.            0 Crematogaster mimosae
```

b.

For this data, a value of 1 in the case_control column indicates that a nest was observed in the acacia tree in question. A value of zero in that said column means that no bird nest was observed in the tree.

c.

Tree Height (height_cm)

Tree height in this data set is a continuous variable that is measured in centimeters. As tree height increases the probability of nest occurrence would likely increase. Taller trees are more structurally complex and therefore have more nesting opportunities, they may also provide better predator avoidance for birds (From Results paragraph 1 and Discussion paragraph 2).

Acacia ant species (ant_species)

Ant species is a categorical variable with four factors. As ant species changes, the probability of nest occurance would likely differ. This is due to each ant species exhibiting a distinct aggression level and mutualistic relationship with the host Acacia tree which affects the level of predator protection a nest in said tree would receive. (Results, paragraphs 1 + 2, Introduction, paragrpah 1, Discussion, paragraph 3)

d.

Model number	Ant Species	Tree Height	Predictor list
1	X	X	all predictors, no interaction (null model)
2	X	X	all predictors, with interaction (full model)

e.

f.

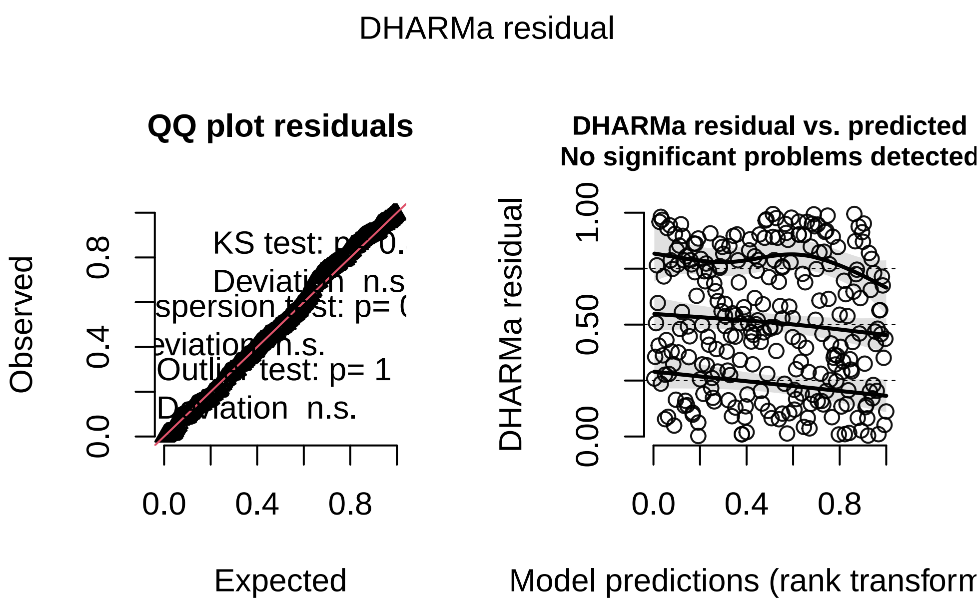
```
# running AIC test to determine best model
AIC(model1, model2)
```

```
       df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) was the model that predicted test_case (whether or not a sample tree had a nest) using the interaction of tree height (height_cm) and species of acacia ant (ant_species)

g.

```r
# plotting simulated residuals from DHARMa package.
plot(
  simulateResiduals(model2)
)
```



Simulated residuals look good, QQ has very little deviation from expected and residual vs predicted looks random with no discernable pattern.

h.

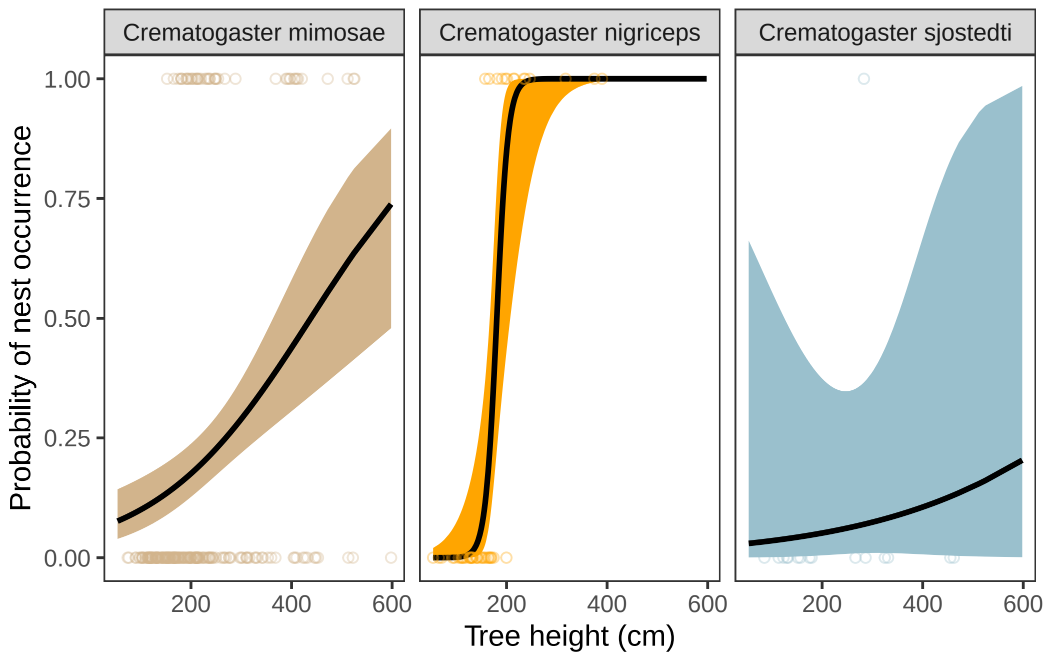
```r
# creating prediction model for model2
preds <- ggpredict(model2,
                   # terms for model, all tree heights all species
                   terms = c("height_cm[all]", "ant_species")) |>
  # changing grouping column's name
  rename(ant_species = group) |>
  arrange(x)
```

```r
# setting custom species colors
sp_cols <- c(
  "AB" = "lightblue3",
  "RRB" = "tan",
  "BBR" = "orange"
)

# creating facet labels from nest_clean
sp_labels <- nests_clean |>
  # choosing correct columns
  distinct(ant_species, sci_name) |>
  deframe()

# base layer: ggplot
ggplot() +
  # first layer: model confidence ribbon
  geom_ribbon(data = preds, # prediction data frame
              aes(x = x,
                  ymin = conf.low, # min by low confidence
                  ymax = conf.high, # max by high confidence
                  fill = ant_species)) + # filling color by group
  # second layer: model probability line
  geom_line(data = preds, # prediction data frame
            aes(x = x,
                y = predicted),
            # coloring line black
            color = "black",
            # making line thicker
            linewidth = 1) +
  # third layer: jitter for underlying data
  geom_jitter(data = nests_clean, # nests_clean data frame
              aes(x = height_cm, # x-axis: tree height
                  y = case_control, # y-axis: nest occurance
                  color = ant_species), # coloring by species
              alpha = 0.35, # making points more opaque
              shape = 21, # changing shape
              height = 0) + # removing vertical jitter
  # seperating by species
  facet_wrap(~ ant_species,
             # labelling with sci_name
             labeller = labeller(ant_species = sp_labels)) +
  # setting colors by species
```

```
  scale_color_manual(values = sp_cols) +
  scale_fill_manual(values  = sp_cols) +
  # relabelling x and y-axes
  labs(
    x = "Tree height (cm)",
    y = "Probability of nest occurrence"
  ) +
  # changing theme
  theme_bw()+
  # removing legend
  theme(legend.position = "none",
        # removing gridlines
        panel.grid = element_blank())
```



i.

Figure 1. Model predictions for bird nest occurrence in acacia trees based on height and ant species Figure displays model predictions for probability of nest occurrence as tree height increases, separated by ant species living on the tree. Line represents exact predictions of model that assumes interaction between species and tree height. Ribbon represents model 95% confidence interval. Species are sorted by color with *C. sjostediti* being tan, *C. mimosae* being

orange, and *C. nigriceps* being blue. Data originally from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322.

j.

```
# creating prediction at 600 cm
ggpredict(model2,
          terms = c("height_cm [600]", "ant_species"))
```

```
# Predicted probabilities of case_control

ant_species: RRB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: BBR

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00

ant_species: AB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99
```

```
# showing summary for baseline
summary(model2)
```

```

Call:
glm(formula = case_control ~ height_cm * ant_species, family = "binomial",
    data = nests_clean)

Coefficients:
```

```
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)             -2.845076   0.434500  -6.548 5.83e-11 ***
height_cm                0.006492   0.001572   4.130 3.63e-05 ***
ant_speciesBBR         -13.128072   5.405092  -2.429  0.01515 *
ant_speciesAB           -0.848334   2.554617  -0.332  0.73983
height_cm:ant_speciesBBR  0.081944   0.030921   2.650  0.00805 **
height_cm:ant_speciesAB  -0.002597   0.008386  -0.310  0.75679
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 286.04  on 269  degrees of freedom
Residual deviance: 225.80  on 264  degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7
```

k.

Ant species and tree height interact to predict the probability of a bird nest in a sample acacia tree. As tree height increases, nest probability increases. Additionally, the effect of height on nest occurrence probability is stronger for *C. nigriceps* ant species trees (odds ratio: 1.092) than *C. mimosae* ant species trees (odds ratio: 1.007) or *C. sjostedti* ant species trees (odds ratio: 1.004). At 600 cm (the highest recorded tree height), nest occurrence probability is 0.74 (95% CI: [0.48, 0.90]) for *C. mimosae*, 1.00 (95% CI:[1.00, 1.00]) for *C. sjostedti*, and 0.20 (95% CI: [0.00, 0.99]). It is important to note that the model predicted no significant interaction between tree height and *C. sjostedti*, accounting for the strange-looking figure and non-precise CI. This makes biological sense as that species does not display defensive characteristics and therefore would not affect whether or not a nest is protected. *C. mimosae* acts as a good reference here because it is a good defender but does not alter the tree, meaning that there is interaction but it is more straightforward; especially when compared to *C. nigriceps* which is an aggressive defender but also alters tree structure to produce denser canopies therefore better defending nests (explaining sharp jump to 1 as height increases).

Problem 3. Problem 3. Affective and exploratory visualizations

a.

- My affective visualization is pretty much a completely new way to display data when compared to my initial exploratory plots. In effect, it is still a sort of scatter plot but

point shape and color matter far more than location. There is also no models or metrics of central tendency as my goal with the affective visualization was not to make something that explained the whole dataset.
- Honestly it is pretty much impossible to see any similarities between my plots and my affective piece, they share no discernible characteristics besides the underlying data. I did separate by location in each case but I did so differently each time.
- Between the two exploratory plots there are some trends, mainly that walk time doesn't change too much on average between locations. There is also a pretty clear negative relationship between time left and total walk time that can be seen without any models.
- The main feedback I got was on improving readability. My visual was hard to understand but with time and explanation it became more legible for my peers. The visual improvements and readability improvements I made, upping line thickness, marking stops, using one line, etc., were all based on this conversation with my peers.

b.

Using walk_minutes as a response and week as the predictor I believe the most interesting analysis we can run is a t-test (or Mann-Whitney U if nonparametric) to see if mean (or median) walk time differs across location.

This is important to my central question of what is effecting my walk time to class, seeing if there is a difference across location could potentially invalidate my previous work on this data frame where I assumed non-dffering central tendancy between the two locations allowing for both locations' data to be used together.

Walk time is a continuous numeric variable.

Location is a categorical variable with 2 levels: Campbell (n = 19) and 387 (n = 13)

c.

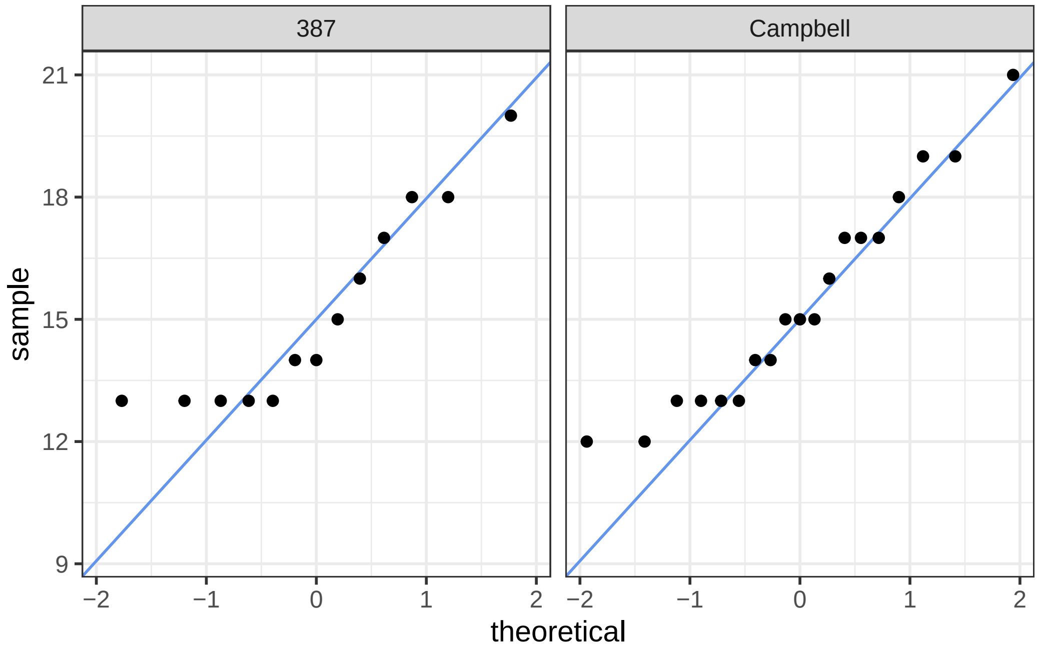
Checking assumptions

- Observations are independent
- Response variable is continuous

Running QQ to test normality

```
# baselayer: ggplot
ggplot(data = personal_data, # using urchins clean data frame
       aes(sample = walk_minutes)) + # y-axis for QQ plot
  # first layer: QQ reference line
  geom_qq_line(color = "cornflowerblue") + # displaying line in blue
```

```
  # second layer: QQ plot
  geom_qq() +
  # creating facets
  facet_wrap(~location) +
  # changing theme
  theme_bw()
```



Not very normal looking, some grouping along line but likely need to use non-parametric due to low sample size.

Running analysis

```
# Mann-Whitney U test
# Walk_minutes response
# Location predictor
wilcox_test(walk_minutes ~ location,
                    data = personal_data) # pulling from personal data
```

```
# A tibble: 1 x 7
  .y.          group1 group2      n1     n2 statistic     p
```

```
* <chr>        <chr>  <chr>    <int> <int>     <dbl> <dbl>
1 walk_minutes 387    Campbell    13    19      118.  0.83
```

Interpretation: Mann-Whitney test, U = 118, p = 0.83, $\alpha = 0.05$

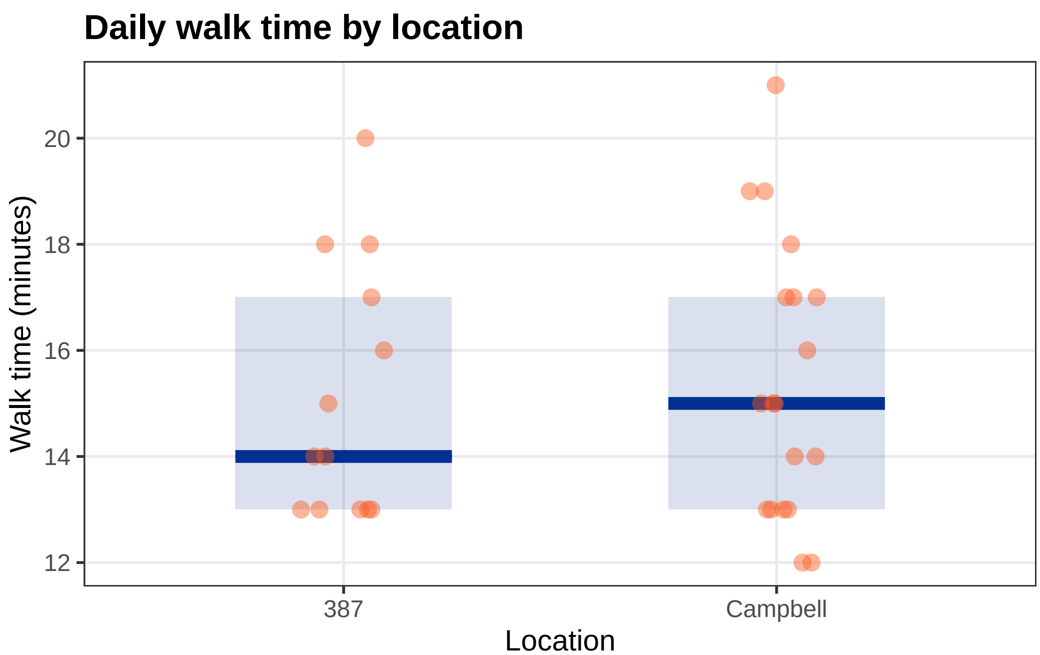
No significant difference in rank distributions.

d.

```r
# creating new location median data frame
location_medians <- personal_data |>
  # grouping by location
  group_by(location) |>
  # summarizing data
  summarise(median_walk = median(walk_minutes), # calculated medians
            q25 = quantile(walk_minutes, 0.25), # calculated CI lower
            q75 = quantile(walk_minutes, 0.75)) |># calculating CI Upper
  # ungrouping
  ungroup()

# base layer: ggplot
ggplot(data = personal_data,
       aes(x = location, # x-axis: location
           y = walk_minutes)) + # y-axis: walk minutes
  # first layer: rectangles for distributions
  geom_rect(data = location_medians,
            # setting rectangle's x max and min
            aes(xmin = as.numeric(factor(location)) - 0.25,
                xmax = as.numeric(factor(location)) + 0.25,
                ymin = q25, # ymin at lower quartile
                ymax = q75, # ymax at upper quartile
                x = NULL,
                y = NULL),
            # filling color
            fill = "#003090",
            # making slightly transparent
            alpha = 0.15) +
  # second layer: cross bars for medians
  geom_crossbar(data = location_medians,
                # setting to display medians for each location
                aes(y = median_walk,
```

```
                    ymin = median_walk,
                    ymax = median_walk),
                # coloring same as rectangle
                color = "#003090",
                width = 0.5, # make bar less wide
                linewidth = 0.9) + # making bar less thick
# third layer: jitter for underlying data
geom_jitter(height = 0, # removing vertical jitter
            width = 0.1, # slight horizontal jitter
            size = 2.5, # making points larger
            alpha = 0.45, # making points transparent
            color = "#FE5A1D") + # changing color
# relabeling title and axes
labs(title = "Daily walk time by location",
     x = "Location",
     y = "Walk time (minutes)") +
# changing theme
theme_bw() +
# further theme adjustments
theme(
  plot.title = element_text(face = "bold"), # bolding title
  panel.grid.minor = element_blank()) # removing minor gridlines
```



e.

Figure 2. No significant difference in walk time distributions across locations. Blue cross bars represent median walk time to class for each location (Campbell n = 19 and 387 n = 13) with the rectangle representing the IQR for each week's observations. Underlying data is shown via the orange points which each represent a single day's walk time. Data collected by Luke Simon between the dates of 2026-04-20 and 2026-06-03.

f.

From the dates of 2026-04-20 and 2026-06-03 there is no difference in daily walk time distributions between building 387 (median = 14, n = 13) and Campbell Hall (median = 15, n = 19) (Mann-Whitney test, $U = 118$, $p = 0.83$, $\alpha = 0.05$). Since our result is not significant we can assume that distributions are nearly identical (as is represented in the above figure) meaning that all previous analysis done and linear models created that do not differentiate between location are still valid. Essentially, walk times to each location are comparable and can be treated as independent observations in the same data set, any differentiation between the two locations would be included for clarity and aesthetic purposes only.